GE SONG

Ph.D. Student in Chemistry, Duke University | Machine Learning for Molecular Simulation & AI for Biophysics
g.song@duke.edu | Personal Website | Google Scholar | GitHub

Ph.D. researcher developing physically grounded machine-learning methods for biomolecular simulation. My work integrates machine learning, molecular mechanics, and quantum chemistry to study biomolecular conformations and enzyme reaction mechanisms. More broadly, I am interested in using AI to capture the high-dimensional, multiscale complexity of living systems and to build computational models that explain and predict complex biological processes, with potential applications in drug discovery, disease treatment, and healthy lifespan extension.

EDUCATION

Duke University, Durham, NC Sep 2023 – Present
Ph.D. in Chemistry | Advisor: Prof. Weitao Yang

Shanghai Jiao Tong University, Shanghai, China Sep 2019 – Jun 2023
B.S. in Bioinformatics | GPA: 89.37/100, Rank: 3/29

SELECTED RESEARCH EXPERIENCE

Reaction Mechanisms in Radical SAM Enzymes 2025 – Present

Duke University | Advisor: Prof. Weitao Yang | Collaboration with Prof. Kenichi Yokoyama

- Investigating radical formation and hydrogen-transfer pathways in a Fe–S cluster enzyme using broken-symmetry DFT, QM/MM, transition-state optimization, kinetic isotope effect analysis, and free-energy calculations.
- Developing machine-learning approaches to accelerate QM/MM free-energy calculations along reaction pathways.

General Machine-Learning Force Field for Biomolecular Simulation 2023 – Present

Duke University | Advisor: Prof. Weitao Yang

- Developing a transferable and scalable machine-learning/molecular-mechanics model for proteins in aqueous environments, with NepoIP/MM representing an initial stage of this broader effort through its treatment of electrostatics effects in the machine-learning model from the MM environment.
- Implemented NepoIP/MM in OpenMM and demonstrated stable biomolecular simulations with agreement to QM/MM reference calculations.

Related outputs: JCTC paper | Open-source code: Interface with OpenMM

Deep-Learning Modeling of Physiological Signals for Sleep Apnea 2021 – 2022

Shanghai Jiao Tong University | Advisor: Prof. Jin Ma

- Adapted a Vision Transformer (ViT) architecture to model multichannel physiological time-series data, including EEG and other polysomnography signals, for obstructive sleep apnea analysis.
- Developed an end-to-end deep-learning workflow for continuous sleep-depth annotation and computer-aided diagnosis, contributing to a publication in npj Digital Medicine.

Force-Field Development for RNA and Modified Proteins 2021 – 2024

Shanghai Jiao Tong University | Advisor: Prof. Haifeng Chen

- Developed and assessed biomolecular force fields for phosphorylated proteins and RNA.
- Contributed to multiple peer-reviewed publications, including first/co-first-author work in JCIM and JCTC.

SELECTED PUBLICATIONS

Song, G.; Yang, W.* NepoIP/MM: Toward Accurate Biomolecular Simulation with a Machine Learning/Molecular Mechanics Model Incorporating Polarization Effects. *J. Chem. Theory Comput. (2025).*

Zhou, S.; Song, G.; Sun, H.; et al. Continuous Sleep Depth Index Annotation with Deep Learning Yields Novel Digital Biomarkers for Sleep Health. *npj Digital Medicine (2025).*

Zheng, X.#; Song, G.#; Li, Z.#; et al. Nonbond Parameter Optimization Improving Simulation of Intrinsically Disordered Phospho-proteins. *J. Chem. Inf. Model. (2025).*

Li, Z.#; Song, G.#; Zhu, J.#; et al. Excited-Ground State Transition of RNA Strand Slippage Mechanism Captured by Base-specific Force Field. *J. Chem. Theory Comput. (2024).*
Song, G.; Zhong, B.; Zhang, B.; et al. Phosphorylation Modification Force Field FB18CMAP Improving Conformation Sampling of Phosphoproteins. *J. Chem. Inf. Model. (2023).*
Additional publications are available on Google Scholar.

AWARDS & PRESENTATIONS

Peter Walter Jeffs Fellowship — Jun 2025

Best Bachelor Thesis, Top 1%, Shanghai Jiao Tong University — Jun 2023

First Prize, Shanghai Challenge Cup Academic Science and Technology Competition — May 2023

Outstanding Graduate, Shanghai Jiao Tong University — May 2023

First Prize, China Undergraduate Life Sciences Contest — Aug 2022

Oral Presentation: “NepoIP/MM: Toward accurate biomolecular simulation with a machine learning/molecular mechanics model incorporating polarization effects” ACS Spring Meeting, Atlanta, GA, Mar 2026.

AI-ASSISTED RESEARCH EXPERIENCE

- Use Claude Science, Codex, and Cursor to support literature organization, scientific coding workflows, data processing, model implementation and optimization, and result visualization.
- Apply AI tools as part of an iterative research workflow while validating outputs through physical reasoning, numerical tests, and reproducible simulation protocols.

TECHNICAL EXPERTISE

While AI agents are becoming increasingly capable, my scientific training and hands-on research experience enable me to use them more effectively in the following areas:

Molecular Simulation: QM/MM; molecular dynamics; free-energy calculations; transition-state search
Machine Learning: E(3)-equivariant neural network; atomistic representation learning